SAT Unit 1.4
System of Two Linear Equation in Two Variables

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Overview

Solving System of the Equation

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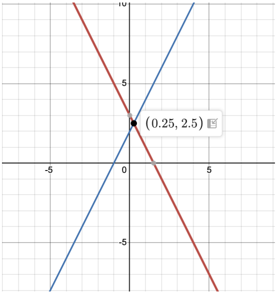
What is a System of Equations?

- A system of equations is a set of two or more equations with the same variables.
- We look for a solution that satisfies all equations simultaneously.
- In two variables (usually x and y), a system of linear equations is:

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

- The solution is the point (x, y) where the two lines intersect.



Solving by Graphing

- Graph both equations on the coordinate plane.
- The point of intersection is the solution.
- This method is visual but not always accurate without graphing tools.

Example:

$$\begin{cases} y = 2x + 1 \\ y = -x + 4 \end{cases} \quad \Rightarrow \quad \text{Intersect at } (1, 3)$$

Solving by Substitution

Steps:

1. Solve one equation for one variable.
2. Substitute that expression into the other equation.
3. Solve for the remaining variable, then back-substitute to find the other.

Example:

$$\begin{cases} y = 3x + 2 \\ 2x + y = 16 \end{cases} \quad \Rightarrow 2x + (3x + 2) = 16$$

Solving by Elimination

Steps:

1. Align both equations.
2. Multiply one or both equations to make coefficients cancel.
3. Add/subtract to eliminate one variable.
4. Solve for one variable, then substitute to find the other.

Example:

$$\begin{cases} 3x + 2y = 16 \\ 5x - 2y = 4 \end{cases} \Rightarrow 8x = 20 \Rightarrow x = 2,\ \ y = 5$$

Question: Solve for y

Solve for y in the system of equations:

$$-x + y = -3.5$$
$$x + 3y = 9.5$$

If (x, y) satisfies the system above, what is the value of y?

Solution

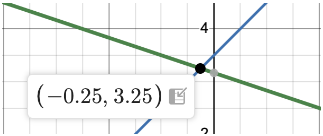
Step 1: Solve for x

- Rewrite the first equation for x:

$$x = y + 3.5$$

- Substitute into the second equation:

$$(y + 3.5) + 3y = 9.5$$



Step 2: Solve for y

- Simplify:

$$4y + 3.5 = 9.5$$

- Divide by 4:

$$y = \frac{6}{4} = 1.5$$

System of Equations — Parametric Point

Given:

$$2x + 3y = 7$$
$$10x + 15y = 35$$

For each real number r, which of the following points lies on the graph of **each** equation in the xy-plane?

(A) $\left(\frac{r}{5} + 7,\ -\frac{r}{5} + 35\right)$
(B) $\left(-\frac{3r}{2} + \frac{7}{2},\ r\right)$
(C) $\left(r,\ \frac{2r}{3} + \frac{7}{3}\right)$
(D) $\left(r,\ -\frac{3r}{2} + \frac{7}{2}\right)$

Solution

Step 1: Notice the system of equations

$$2x + 3y = 7$$
$$10x + 15y = 35$$

- Second equation is a multiple of the first: $5(2x + 3y) = 10x + 15y = 35$.
- Therefore, both equations are equivalent — they represent the same line.

Step 2: Pick a parametric form

- Solve $2x + 3y = 7$ for x:
$$x = -\frac{3}{2}y + \frac{7}{2}$$
- Let $y = r$, then:
$$x = -\frac{3}{2}r + \frac{7}{2}$$
- So the point on the line is $\left(-\frac{3}{2}r + \frac{7}{2}, r\right)$

Final Answer: $\boxed{\text{B.}}$

Another Solution — Answer: B

Step 1: Observe the system of equations

$$2x + 3y = 7$$
$$10x + 15y = 35$$

- Second equation is a multiple of the first:

$$5(2x + 3y) = 10x + 15y = 35$$

- So both equations describe the same line.

Step 2: Plug the coordinates of each choice into the first equation

- Choice (B): $x = -\frac{3r}{2} + \frac{7}{2}, \quad y = r$
- Plug into: $2x + 3y = 7$

$$2\left(-\frac{3r}{2} + \frac{7}{2}\right) + 3r = -3r + 7 + 3r = 7$$

- True! So this point lies on the line.

Answer: $\boxed{\text{B}}$

Types of Solutions

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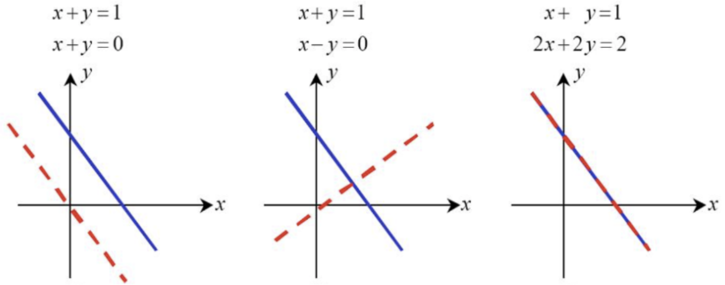
Types of Solutions

- **One solution:** The lines intersect at one point.
- **No solution:** The lines are parallel and never intersect.
- **Infinitely many solutions:** The lines are identical.



Question: Types of Solutions

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

$$\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$$
$$\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$$

Solution

Step 1: Simplify both equations

First equation:

$$\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y \Rightarrow \frac{3}{2}y + \frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} \Rightarrow 3y - \frac{1}{4}x = \frac{2}{3}$$

Second equation:

$$\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2} \Rightarrow \frac{1}{2}x + \frac{3}{2} - \frac{9}{2} = py \Rightarrow \frac{1}{2}x - 3 = py$$

Step 2: Write both in standard form

From the first:

$$\frac{1}{4}x - 3y = -\frac{2}{3}$$

From the second:

$$\frac{1}{2}x - py = 3$$

Step 3: No solution condition

To have no solution, the lines must be parallel — same slope, different intercepts.

$$\frac{\frac{1}{4}}{-3} = \frac{\frac{1}{2}}{-p} \Rightarrow \frac{1}{4} \cdot (-p) = \frac{1}{2} \cdot (-3) \Rightarrow -\frac{p}{4} = -\frac{3}{2} \Rightarrow \frac{p}{4} = \frac{3}{2} \Rightarrow p = 6$$

System of Equations — No Solution

Given:

$$4x - 6y = 10y + 2$$
$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

Solution

Step 1: Rearrange both equations

- Move all terms to one side in the first equation:

$$4x - 6y - 10y = 2 \Rightarrow 4x - 16y = 2$$

- Rearranged second equation:

$$ty = \frac{1}{2} + 2x \Rightarrow -2x + ty = \frac{1}{2}$$

Step 2: Identify slope condition for no solution

- Coefficients must give the same slope:

$$\frac{4}{-16} = \frac{-2}{t} \Rightarrow -\frac{1}{4} = \frac{-2}{t} \Rightarrow t = 8$$

Final Answer: $\boxed{t = 8}$

Real World Application

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Real-World Application

Problem: At a concert, 2 adult tickets and 3 student tickets cost \$54. 3 adult tickets and 2 student tickets cost \$56. What is the price of each ticket?

Let:

- x: adult ticket price
- y: student ticket price

$$\begin{cases} 2x + 3y = 54 \\ 3x + 2y = 56 \end{cases}$$

Practices

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Question: Infinitely Many Solutions – System of Equations

Consider the following system of equations:

$$mx - 6y = 10$$
$$2x - ny = 5$$

In the given system, m and n are constants. The system has **infinitely many solutions**. What is the value of $\frac{m}{n}$?

A. $\frac{1}{12}$

B. $\frac{1}{3}$

C. $\frac{4}{3}$

D. 3

Answer: System of the Equations

To compare, write Equation (2) in the form of Equation (1) by multiplying both sides by a constant k:

$$k(2x - ny) = k \cdot 5 \Rightarrow 2kx - kny = 5k$$

Compare with:

$$mx - 6y = 10$$

Matching coefficients:

$$2k = m, \quad kn = 6, \quad 5k = 10 \Rightarrow k = 2$$

Substitute $k = 2$:

$$m = 2k = 4, \quad n = \frac{6}{k} = 3 \Rightarrow \frac{m}{n} = \frac{4}{3}$$

$$\boxed{\text{Correct Answer: C. } \frac{4}{3}}$$

Question: No Solution – System of Equations with Fractions

Consider the system of equations below:

$$\frac{3}{10}y + \frac{b}{3}x = \frac{3}{4} - \frac{1}{5}y$$
$$\frac{3}{8}x + \frac{3}{5} = -\frac{1}{4}y + \frac{8}{5}$$

In the given system, b is a constant. If the system has **no solution**, what is the value of b?

Answer: No Solution – System of Equations

Step 1: Rewrite both equations in standard form

- First equation:

$$\frac{3}{10}y + \frac{b}{3}x = \frac{3}{4} - \frac{1}{5}y \Rightarrow \left(\frac{3}{10} + \frac{1}{5}\right)y + \frac{b}{3}x = \frac{3}{4} \Rightarrow \frac{1}{2}y + \frac{b}{3}x = \frac{3}{4}$$

- Second equation:

$$\frac{3}{8}x + \frac{3}{5} = -\frac{1}{4}y + \frac{8}{5} \Rightarrow \frac{3}{8}x + \frac{3}{5} - \frac{8}{5} = -\frac{1}{4}y \Rightarrow \frac{3}{8}x - 1 = -\frac{1}{4}y \Rightarrow \frac{1}{4}y + \frac{3}{8}x = 1$$

Step 2: Standard forms for comparison

$$\text{First: } \frac{b}{3}x + \frac{1}{2}y = \frac{3}{4} \qquad \text{Second: } \frac{3}{8}x + \frac{1}{4}y = 1$$

Step 3: No solution requires same slope, different intercepts

Make slopes equal:

$$\frac{b}{3} : \frac{1}{2} = \frac{3}{8} : \frac{1}{4} \Rightarrow \frac{b/3}{1/2} = \frac{3/8}{1/4} \Rightarrow \frac{2b}{3} = \frac{3}{2} \Rightarrow b = \frac{9}{4}$$

Correct Answer: $\boxed{b = \frac{9}{4}}$

System of Equations — Parametric Point

Given:

$$3x - 2y = 6$$

$$6x - 4y = 12$$

For each real number r, which of the following points lies on the graph of **each** equation in the xy-plane?

A. $\left(r, \frac{3r}{2} - 3\right)$
B. $\left(\frac{2r-6}{3}, r\right)$
C. $\left(r, \frac{3r}{2} - 6\right)$
D. $\left(\frac{r-6}{2}, r\right)$

Answer: Parametric Point

Step 1: Note the system of equations

$$3x - 2y = 6 \quad \text{and} \quad 6x - 4y = 12$$

These are equivalent (the second is just the first multiplied by 2), so any point that satisfies one will satisfy the other.

Step 2: Check point from Option A:

$$(x, y) = \left(r, \frac{3r}{2} - 3\right)$$

Substitute into the first equation:

$$3r - 2\left(\frac{3r}{2} - 3\right) = 3r - (3r - 6) = 6 \Rightarrow \text{True!}$$

So this point lies on both lines for any real r.

Correct Answer: $\boxed{\textbf{A}}$

System of Equations — Parametric Point

Given:

$$4x + y = 10$$

$$12x + 3y = 30$$

For each real number r, which of the following points lies on the graph of **both** equations in the xy-plane?

A. $(r, -4r + 10)$

B. $\left(\frac{r-10}{4}, r\right)$

C. $(r, 4r + 10)$

D. $\left(r, \frac{10-r}{4}\right)$

Answer: Parametric Point

Step 1: Recognize proportional system. The second equation is exactly 3 times the first:

$$12x + 3y = 3(4x + y) = 30$$

So any point satisfying the first also satisfies the second — they are the same line.

Step 2: Try Option A:

$$(x, y) = (r, -4r + 10)$$

Substitute into the first equation:

$$4r + (-4r + 10) = 10 \Rightarrow 10 = 10 \quad \text{True!}$$

Correct Answer: $\boxed{\textbf{A}}$

System of Equations — No Solution (Fractions)

Given the following system of equations:

$$\frac{4}{3} + \frac{5}{4}x = 28y + \frac{5}{8}x$$

$$my = \frac{1}{2}(5x - 8)$$

In the given system of equations, m is a constant. If the system has **no solution**, what is the value of m?

Answer: System of Equations

Correct Answer: $\boxed{m = 112}$

Thank You!

We appreciate your time and focus.

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